PRIMARY TEACHING NOTES SERIES

ICT in the Primary School

How to use phones, screens, audio, pictures and simple digital tools to teach young children better — even in a classroom with no computer lab and no internet.

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These are study and practice notes for primary teachers, written in simple English. They cover one topic only — Information and Communication Technology (ICT) in primary classes — and follow the themes taught in teacher induction training: what ICT is, its benefits and limits, everyday tools, subject-wise use, lesson planning, offline methods, safety and screen time.

What ICT Means in a Primary School

Many teachers hear the words ICT and think of a computer lab. Most primary schools do not have one — and they do not need one to use ICT well. In a primary classroom, ICT simply means using technology to help children learn.

> **Plain idea:** ICT stands for Information and Communication Technology. In school it means any tool that shows, plays, records, practises or shares information for learning — a phone, a television, a speaker, a laptop, a projector, or even a simple audio lesson.

Four jobs ICT does in class

- **Show.** Pictures, videos and models make things visible that children cannot see in real life — the inside of a plant, a faraway city, or how a letter is formed.
- **Play.** Songs, rhymes, stories and sounds bring language and rhythm into the room in the correct voice and pronunciation.
- **Practise.** Quizzes, flashcards and learning games give every child many turns, with instant right-or-wrong feedback.
- **Record and share.** A photo of good work, a recorded reading, or a class result shared with parents keeps a record and builds pride.

ICT is a servant, not the teacher

Technology does not teach by itself. The teacher still chooses the objective, asks the questions, watches the children and corrects mistakes. A video played without questions is entertainment. The same video, paused at the right moments with questions before, during and after, becomes a lesson.

Start small. One tool, used well for ten minutes, teaches more than five tools used in confusion for a whole period.

Why ICT Helps Young Learners — and Its Limits

Young children learn through their senses. They need to see, hear, touch and do. ICT is powerful at the primary level because it feeds sight and hearing together, and it lets a child repeat an activity as many times as needed without shame.

Benefits for primary children

- **Makes the abstract concrete.** Fractions, the water cycle and number bonds are hard to imagine; a clear animation or picture makes them plain.
- **Correct models.** Children hear correct English and Urdu pronunciation, letter sounds and number names every time, in the same clear voice.
- **Motivation and attention.** Colour, movement and sound hold the attention of young learners — especially at the start of a new topic.
- **Practice without fear.** A child can try, fail and try again on a game or quiz without the whole class watching.
- **Support for different learners.** Slow learners can replay; fast learners can move ahead; a child who missed school can catch up with the same material.

Limits a wise teacher remembers

- Screens cannot replace talking, handwriting, counting real objects or playing outside.
- Too much screen time tires eyes and shortens attention for normal lessons.
- Content from the internet is not always correct, suitable or in the child's language — the teacher must check it first.
- Electricity, batteries and signal can fail at any moment. A good ICT lesson always has a non-digital backup.

> **Golden rule:** Use ICT for the part of the lesson it does best — showing, modelling or practising — then return to books, slates, objects and discussion for the rest.

The ICT Tools a Primary Teacher Really Has

You do not need the newest equipment. Most government primary teachers already have, or can borrow, the tools in this table. Learn one tool fully before adding another.

Tool	Best classroom uses
Teacher's smartphone	Showing pictures and short clips, playing audio, taking photos of work, timer, torch, recording children reading aloud.
TV or large screen	Whole-class lessons, rhymes and stories, showing one clear picture or model to everyone at once.
Laptop or tablet	Preparing lessons, offline learning apps and games, quizzes, showing slides, keeping class records.
Bluetooth speaker	Rhymes, listening exercises, stories and pronunciation practice that the whole class can hear clearly.
Projector (if available)	Big, bright display for the whole class; useful for shared reading and step-by-step worked examples.
Radio / audio lessons	Listening skills, songs and stories — cheap, reliable and works without internet.
Memory card / USB	Carrying downloaded videos, songs and pictures from home or town into a school with no connection.

Teacher tools behind the scenes

ICT also helps the teacher, not only the children: preparing worksheets, keeping attendance and result records, updating school data, and communicating with parents and the education department. Doing this work neatly on a phone or computer saves hours every month and keeps records safe.

> **Start where you are:** If you have only a phone and a speaker, you already have enough for audio stories, rhymes, pictures and recorded reading. Begin there.

ICT in Every Subject

ICT earns its place when it fits the subject. The same tool does a different job in each lesson.

Languages — English, Urdu and Pashto

- Play rhymes and songs for sounds, rhythm and new words; children sing, then read the words.
- Use audio stories for listening, then ask questions and let children retell the story in their own words.
- Show a big picture and build oral sentences about it before any writing begins.
- Record a child reading aloud; playing it back helps the child hear and correct their own mistakes.

Mathematics

- Use short animations for counting, place value, shapes, fractions and measurement.
- Run quick mental-maths quizzes and games for tables and number bonds — then practise the same sums on slates.
- Photograph real objects — stones, sticks, fruit — and count, sort and group them on screen together.

Science and social studies

- Show what cannot be brought into class: animals, weather, planets, maps, historical places and other countries.
- Use a short clip before a real activity — for example, watch germination, then plant seeds in the school ground.

Art, physical activity and values

Step-by-step drawing videos, action songs and movement breaks, and short stories about honesty, kindness and cleanliness all work well — as long as children then draw, move and discuss themselves.

Planning an ICT Lesson, Step by Step

An ICT lesson is planned like any good lesson. The objective comes first; the technology comes second. If you cannot say what children will learn, no video or app will save the lesson.

1. **Write the objective.** One clear aim, for example: by the end, children can name five parts of a plant.
2. **Choose the smallest tool that fits.** A single picture may do the job better than a long video.
3. **Check everything before class.** Watch the full clip, test the app, charge the device, download the file. Never show unchecked content to children.
4. **Plan questions, not just playing.** Write one question before viewing, two during, and one after. Pause the clip to ask them.
5. **Plan the follow-up activity.** After watching, children must do something: label a diagram, write sentences, sort objects or discuss in pairs.
6. **Keep a backup ready.** If power or signal fails, teach the same objective with the board, a chart or real objects. The lesson must never collapse with the electricity.
7. **Check learning at the end.** A two-minute oral quiz, exit question or slate work shows whether the objective was met.

> **The 10-minute guide:** In a normal period, about ten minutes of screen or audio is usually enough for young children. The rest of the period belongs to talking, doing, reading and writing.

A common mistake to avoid

Do not build the whole lesson around one long video. Children become silent watchers, the period ends, and nothing has been practised or written. Short input, active children — that is the pattern that works.

ICT Without Internet: Low-Resource Classrooms

Many primary schools have weak signal, no Wi-Fi and frequent load-shedding. ICT can still work every day if the teacher plans for offline use.

Download first, teach later

- Download videos, songs, pictures and apps at home or in town, where the connection is better. Save them on the phone, a memory card or a USB.
- Keep a simple class folder system: one folder per subject, one sub-folder per topic, so any file is found in seconds.
- Share good resources teacher to teacher by Bluetooth or memory card — one download can serve the whole school.

One device, whole class

- **Teacher-led showing:** one phone or laptop connected to a TV or speaker is enough for stories, models and quizzes.
- **Rotation:** small groups take short turns on one tablet while others do slate or book work, then groups exchange.
- **Peer helpers:** a confident older child can operate the device under the teacher's eye, freeing the teacher to teach.

Power-smart habits

Charge devices the night before. Keep brightness moderate. Close unused apps. Where the school has solar charging, agree a simple rota so the teaching device is charged first. A printed chart of the same picture is the final backup — keep making charts; they never run out of battery.

Safety, Screen Time and Care of Equipment

Young children cannot judge what is safe or suitable online. In primary school, all ICT use is chosen, started and supervised by the teacher. Children do not browse freely.

Simple safety rules for primary children

- Only open what the teacher has opened or approved.
- Never share your name, photo, school or home details with strangers online.
- If something strange, frightening or rude appears, stop and tell the teacher at once.
- Be kind in every message and comment — the same manners as in the classroom.
- Ask before taking or sharing anyone's photograph.

Healthy screen habits

- Keep screen segments short — about ten minutes at a time for young children — with looking-away and movement breaks.
- Sit children at a sensible distance from a large screen, in good light, not in a dark room.
- Balance every screen activity with real activity: talk, books, objects, drawing, writing and outdoor play.

Care of equipment

Clean, dry hands before touching devices. No food or water near them. Carry with two hands, store in one fixed safe place, and charge under a teacher's eye. A small class ICT monitor can help with these routines, but responsibility stays with the teacher. Report faults early — a small problem fixed in time saves the device.

> **Teacher's promise:** Every clip watched full before class, every app tested, every session supervised. That one habit prevents almost every ICT problem in a primary school.

ICT Checklist and Quick Revision

Use this checklist before and after any ICT lesson. Aim for a yes on every line.

Before the lesson

- My objective is written, and the technology clearly serves it.
- I have watched or tested the full content myself.
- The file is downloaded and the device is charged.
- My questions — before, during and after — are ready.
- A non-digital backup activity is ready if power or signal fails.

During and after the lesson

- Screen time stayed short, and children talked, did and wrote afterwards.
- Every child could see and hear clearly.
- I checked learning at the end, not just enjoyment.
- Equipment was handled carefully and stored safely.

Quick revision — answer in one line each:

1. What does ICT stand for, in your own words?
2. Name the four jobs ICT does in class.
3. Give two benefits and two limits of ICT for young children.
4. You have only a phone and a speaker. Plan one language activity.
5. Write the seven steps of planning an ICT lesson.
6. How will you teach with ICT on a day with no internet and no power?
7. State three safety rules you will teach your class.

Remember: Objective first, technology second. Show less, ask more, and always follow the screen with real work by the children.